The Effect of Speech Enhancement on Whisper Speech Recognition

ELEC5305 Acoustics, Speech and Signal Processing

Audio Processing Individual Project Report

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1. Introduction

Speech enhancement is often placed before automatic speech recognition (ASR) to reduce additive noise. However, enhancement methods are usually designed for signal quality or human listening. They are not designed for the internal features of a fixed recogniser. Noise removal may help recognition, have little effect, or add new distortions. This project asks whether front-end enhancement changes the word error rate (WER) of a fixed Whisper model. It tests different noise types and input signal-to-noise ratios (SNRs).

The comparison includes noisy speech with no enhancement, power spectral subtraction, Wiener filtering, and pretrained DeepFilterNet3. Clean-speech recognition is used as a reference, and no neural model is trained. I did not assume that enhancement would improve recognition, because better SNR or STOI may still be accompanied by worse WER. Compared with the earlier five-utterance experiment, this study adds STOI, Whisper log-Mel distance, corpus WER, error counts, runtime, and a larger test set.

I built a MATLAB-Python pipeline so that every method receives the same saved mixture. Development and test speakers are separate, parameters are chosen before the test run, and every utterance result is saved. I also changed only the spectral-subtraction coefficient to test how stronger noise removal affects residual noise, speech distortion, and WER.

2. Literature Review

Boll's spectral-subtraction method estimates a noise spectrum during non-speech and subtracts it from the noisy short-time spectrum [1]. It is computationally light but can create isolated residual components and speech distortion when the noise estimate is inaccurate. Wiener filtering instead applies a smooth frequency-dependent minimum-mean-square-error gain derived from estimated speech and noise power [2]. Both methods depend on quasi-stationary noise estimates and retain the noisy phase in the implementations used here.

Signal metrics answer different questions. Output SNR measures sample-by-sample error against an aligned clean reference. STOI estimates the intelligibility of noisy and time-frequency weighted speech. It agrees with many listening tests, but it is not designed for ASR [3]. Therefore, neither metric guarantees that Whisper will keep the speech cues it needs.

Whisper was trained on 680,000 hours of diverse weakly supervised audio and is intended to transfer to new tasks without task-specific fine-tuning [6]. This does not mean that every enhancement front end will help. DeepFilterNet uses ERB-band gains and deep filtering and is designed for real-time enhancement [7]. Previous ASR research shows that enhancement artefacts can harm a recogniser even when normal enhancement scores improve [8], [9]. For this reason, Whisper is kept fixed and WER is measured directly rather than treating cleaner audio as proof of better recognition.

Plantinga et al. [8] trained a single-channel enhancer with a perceptual loss from a pretrained acoustic model. On Voicebank+DEMAND, their ASR-aware system reported 2.80% WER. This was more than 20% relatively lower than joint training and 14% lower than distortion-independent mask training. The result shows that enhancement can help ASR when the training objective protects recognition-relevant information.

Braun and Gamper [9] studied a real-time neural noise suppressor. They varied spectral loss terms and tested extra losses from pretrained MOS, WER, ASR and sound-event models. Their results showed a trade-off between noise reduction and speech distortion. The extra pretrained networks did not significantly outperform a strong spectral loss.

This project uses these studies as motivation but tests a different setting. It does not train or adapt an enhancer or recogniser. Two classical methods and pretrained DeepFilterNet3 are placed before fixed Whisper base.en. The comparison covers three noise types, three SNRs and one change in spectral-subtraction strength. The experiment therefore tests front-end compatibility rather than ASR-aware enhancement training. Any agreement with [8], [9] is limited to this 50-utterance read-speech test.

LibriSpeech provides aligned 16 kHz read English speech [4], while DEMAND provides real environmental noise recordings [5]. Controlled mixtures from these datasets make it possible to compare the front ends with the same inputs. The official Whisper 80-bin log-Mel features are also compared with the matched clean reference. This distance is used only to help explain the results; a smaller value does not always mean fewer recognition errors.

3. Methods

3.1 Data and experimental design

A ten-utterance Mini LibriSpeech pilot first checked the five-condition workflow at 0 dB office noise. The formal experiment used a project-specific subset of LibriSpeech dev-clean-2, not the official LibriSpeech test set. Five development utterances from five speakers were used only for parameter selection. The independent test set contained 50 utterances from 21 speakers; no speaker appeared in both sets. Each test utterance was mixed with white, DEMAND office and DEMAND street-traffic noise at 0, 5 and 10 dB. Four primary processing conditions produced 1,800 noisy/enhanced tests, plus 50 clean recognitions. The development-selected spectral modification added 450 test outputs.

Component	Fixed setting
Audio	Mono, 16 kHz; exact length retained
Development / test	5 utterances / 50 utterances; speaker-disjoint
Noise and SNR	White, DEMAND office, DEMAND traffic; 0, 5, 10 dB
Classical STFT	512 samples; 128-sample hop; square-root periodic Hann
Noise prefix	1.0 s, excluded from recognition and all metrics
Whisper	base.en; English; temperature 0; beam size 5; FP16 off
Randomness	Saved mixture seed, channel/start and common scaling per row

3.2 Mixture construction and alignment

For clean speech s[n] and an extracted noise segment v[n], the noise gain was g = sqrt(Ps /(Pv·10^(SNR/10))). A common factor c = min(1, 0.95/max|[s; s+gv]|) was then applied to both the mixture and clean evaluation reference. This prevents clipping without changing SNR. A one-second noise-only prefix was prepended solely for noise estimation. Stored input SNR was recomputed after writing; the maximum absolute test error was 4.6×10^{-6} dB. All 2,250 test waveforms matched the reference length and had an estimated lag of zero samples.

3.3 Enhancement methods

Both classical methods used the same complex STFT X(k,m), window, hop and prefix-derived noise power estimate Pn(k). Spectral subtraction formed $\hat{P}s = \max(|X|^2 - \alpha Pn, 0.02Pn)$ and used gain G = min(sqrt($\hat{P}s/|X|^2$),1). The original method fixed α=1.0. Wiener filtering used $\hat{P}s = \max(|X|^2 - Pn, 0)$ and $G=\hat{P}s/(\hat{P}s+Pn)$. The noisy phase was retained and perfect-overlap synthesis divided by the accumulated squared-window weights. Relative bypass reconstruction error was below 2.2×10^{-16}.

DeepFilterNet v0.5.6 used its embedded pretrained DeepFilterNet3 model. Audio was resampled 16→48→16 kHz with polyphase filtering. The official delay-compensation option was enabled; one second of temporary tail padding was removed after inference. No clean reference entered any enhancement method.

3.4 Recognition and metrics

Whisper base.en was fixed for every condition. Decoding used language=en, task=transcribe, temperature=0, beam size=5, condition_on_previous_text=false and the same thresholds throughout. EnglishTextNormalizer was applied to reference and hypothesis text before jiwer alignment. Corpus WER was (S+D+I)/N using total substitutions, deletions, insertions and reference words. ΔWER is enhanced WER minus the matched noisy WER in percentage points; negative values indicate improvement. Overall uncertainty was estimated with 10,000 paired bootstrap resamples clustered by utterance.

The additional metrics were output SNR, SNR improvement, STOI, and real-time factor (RTF). I also measured the distance between official Whisper log-Mel features and the matched clean reference. Only valid speech frames were used. The log-Mel distance was used to help explain the results, not to choose a method or prove a cause.

3.5 Development-only modification

The only modification was spectral over-subtraction strength. Development candidates α∈{0.75,1.00,1.25,1.50,2.00} kept every other setting fixed. The pre-declared rule selected lowest corpus WER, then highest mean STOI, then proximity to α=1.0. α=1.50 achieved development WER

13.91%, compared with 14.43% for α=1.00 and 14.64% for no enhancement. It was therefore frozen before test evaluation.

4. Results

4.1 Pilot and integrity checks

The completed ten-utterance pilot contained 258 reference words. Clean, noisy, spectral-subtraction, Wiener and DeepFilterNet3 WERs were 6.59%, 5.81%, 5.43%, 5.43% and 4.65%, respectively. Because the noisy pilot unexpectedly outperformed clean speech and the sample was small, it was used only as a pipeline check. The formal run completed all 450 mixtures, 2,300 Whisper inputs and 2,250 waveform evaluations. No missing audio, duplicate recognition key, alignment error or non-finite sample was found.

4.2 Recognition performance

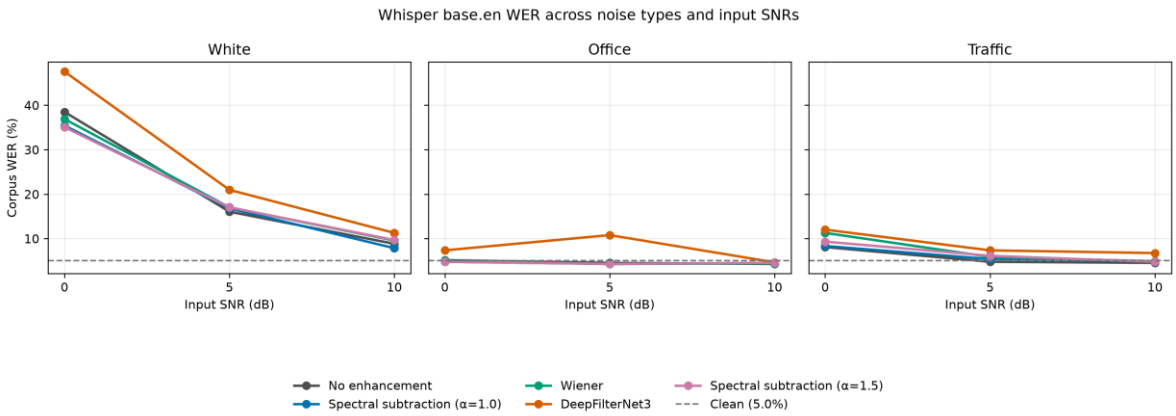


Figure 1. Corpus WER by noise type, input SNR and front-end method. The dashed line is clean-speech WER.

Clean test WER was 5.02%. Without enhancement, overall noisy WER was 10.53% (598 substitutions, 59 deletions and 116 insertions over 7,344 reference words). White noise was most difficult: unenhanced WER fell from 38.48% at 0 dB to 8.82% at 10 dB. Office and traffic conditions were much closer to the clean reference, showing that nominal SNR alone did not predict ASR difficulty.

Method	WER (%)	S	D	I	ΔWER (pp)	95% CI (pp)
No enhancement	10.53	598	59	116	0.00	—
Spectral subtraction α=1.0	10.25	590	55	108	−0.27	[−0.90, 0.34]
Wiener	11.03	623	57	130	+0.50	[−0.18, 1.24]
DeepFilterNet3	14.30	761	71	218	+3.77	[2.14, 5.79]
Spectral subtraction α=1.5	10.62	620	49	111	+0.10	[−0.53, 0.75]

Original spectral subtraction gave the lowest overall WER, but its −0.27-point change was small and its clustered interval included zero. It helped most at white 0 dB (−3.06 points) and white 10 dB (−0.98), while slightly worsening all traffic conditions. Wiener filtering improved white 0 dB by 1.59 points but was worse overall. DeepFilterNet3 increased WER in every noise/SNR cell, with the largest change at white 0 dB (+9.07 points).

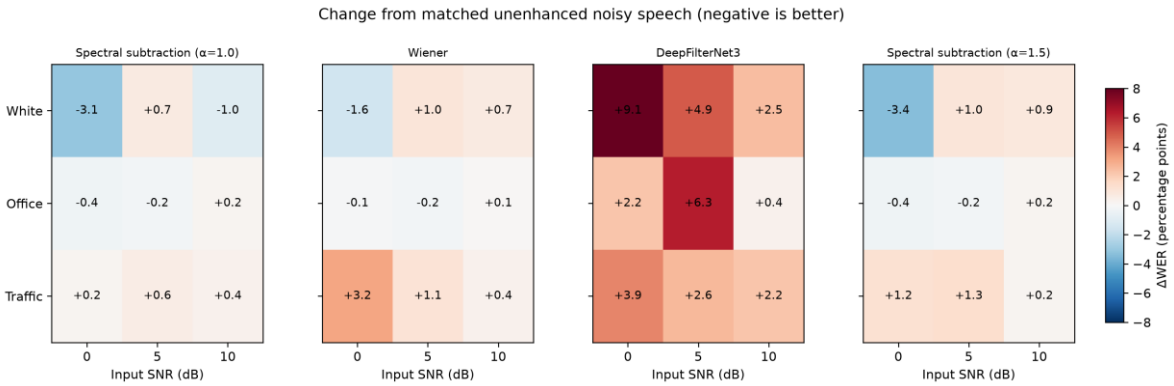


Figure 2. ΔWER relative to the matched unenhanced condition. Negative values indicate improvement.

4.3 Signal metrics versus WER

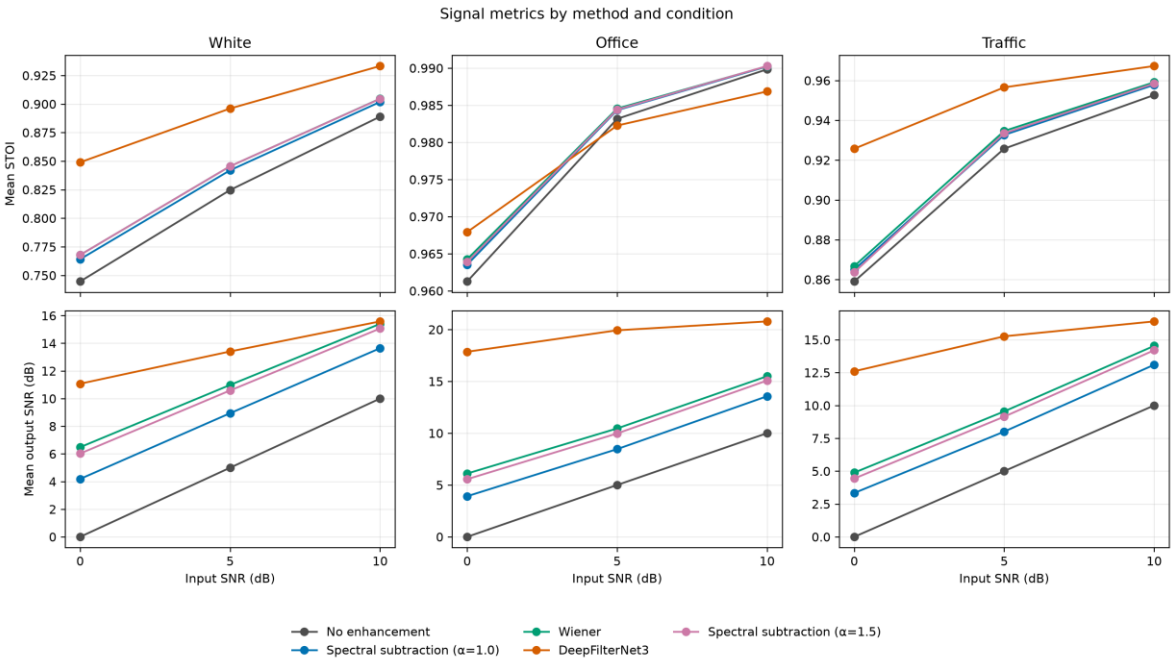


Figure 3. Mean STOI and output SNR. DeepFilterNet3 generally produced the strongest signal-level scores.

Every enhancer increased mean output SNR. DeepFilterNet3 produced about 11-21 dB output SNR. It also gave the highest STOI for white and traffic noise. However, its overall WER was 35.8% higher than the unenhanced result. For office noise at 5 and 10 dB, its STOI was 0.9823 and 0.9869. The unenhanced values were slightly higher at 0.9832 and 0.9898. These results show that better signal scores did not guarantee better Whisper results.

Across the 45 method, noise, and SNR averages, log-Mel MSE and WER had Pearson r=0.67. This shows a useful relationship, but it was not exact. DeepFilterNet3 sometimes moved the average features closer to clean while WER still increased. Whisper may therefore be sensitive to specific distortions that are hidden by an average distance.

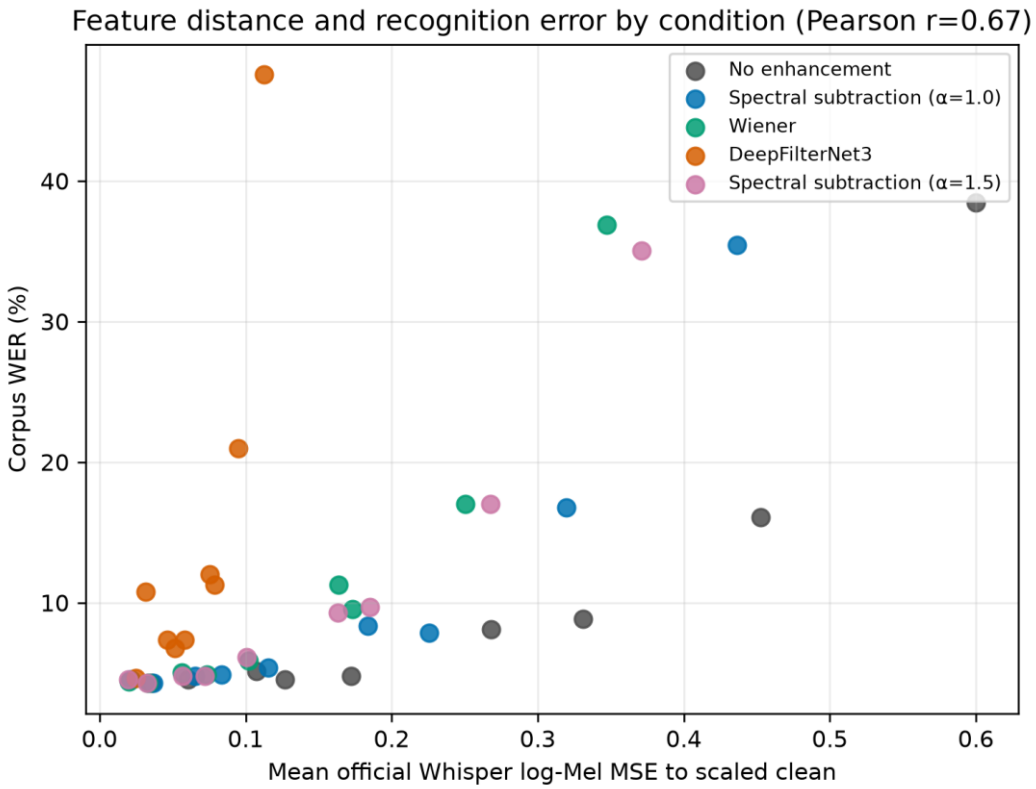


Figure 4. Mean official Whisper log-Mel distance versus corpus WER for each condition.

4.4 Modification and runtime

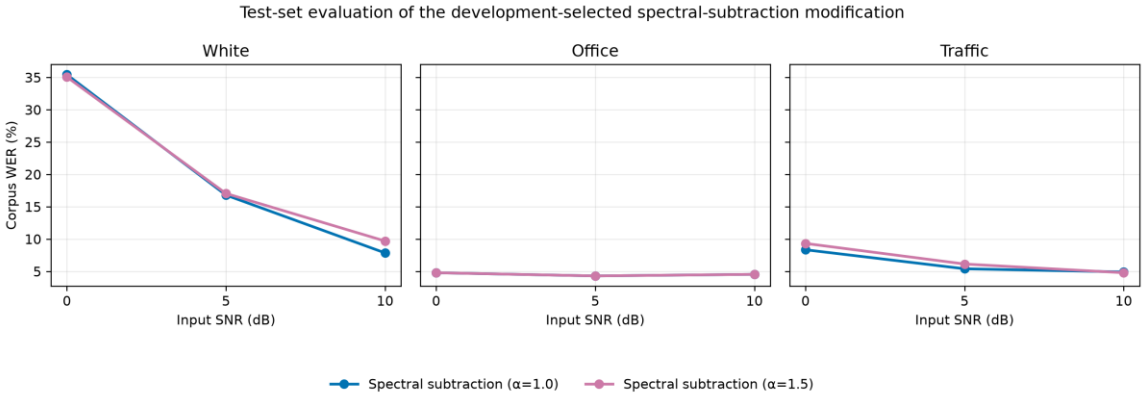


Figure 5. Independent test comparison of original α=1.0 and development-selected α=1.5 spectral subtraction.

The modification did not generalise. Alpha=1.5 was the best setting on the development set. It also slightly improved white noise at 0 dB, with 35.05% WER instead of 35.42%. However, it was worse at white noise at 10 dB and in all traffic conditions. Its overall WER was 10.62%, compared with 10.25% for alpha=1.0. Stronger subtraction improved some signal metrics, but the WER effect

changed with noise and SNR.

The median enhancement RTF was 0.0056 for both classical methods and 0.1457 for DeepFilterNet3. Whisper RTF was about 0.06-0.11 on the experiment computer. One DeepFilterNet process stalled and produced an RTF of 313. This raw value is kept in the CSV. The median is used to describe normal processing speed.

4.5 Success and failure cases

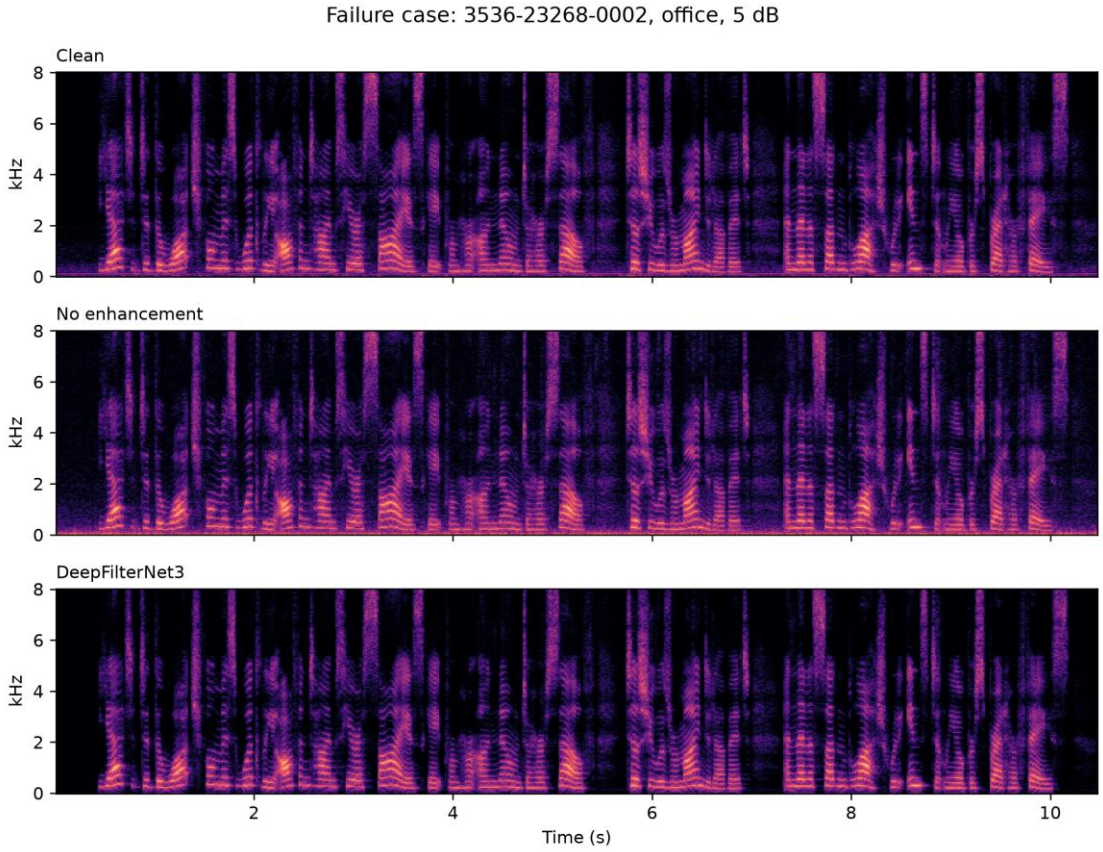


Figure 6. A failure case where DeepFilterNet3 produced a cleaner-looking spectrum but triggered repeated Whisper insertions.

For utterance 3536-23268-0002 in office noise at 5 dB, the reference ended with "then a sudden blush would instantly overspread her face." The unenhanced transcript matched the reference exactly. The DeepFilterNet3 transcript kept this passage but then added "Okay" and repeated "All right." Its 51 errors comprised 1 substitution, 0 deletions and 50 insertions. This gave 159.38% utterance WER; insertions can make WER exceed 100%. DeepFilterNet3 raised output SNR from 5.00 to 19.97 dB, while STOI fell from 0.9746 to 0.9692 and log-Mel MSE rose from 0.0209 to 0.0348. Removing this matched noisy/enhanced pair reduced the overall DeepFilterNet3 penalty from 3.77 to 3.09 percentage points. The method therefore remained worse overall.

The success case showed the opposite pattern. In white noise at 0 dB, spectral subtraction reduced the errors from nine to two. The project results include the audio and transcripts for both cases.

5. Discussion

Enhancement helped in severe white noise, where the stationary noise was close to the estimate from the one-second prefix. In office noise, spectral subtraction and Wiener filtering changed WER by less than half a percentage point at every SNR. DeepFilterNet3 behaved differently. It increased WER by 2.21 points at 0 dB and 6.25 points at 5 dB, despite improving output SNR. Its increase at 10 dB was 0.37 points.

All tested enhancers were neutral or harmful in traffic noise. For the two classical methods, the fixed initial noise estimate may not track the changing traffic noise. This explanation does not directly apply to DeepFilterNet3. Its worse WER may reflect processing distortion or a mismatch between its output and Whisper's expected input, but this experiment did not isolate the mechanism.

The signal metrics and WER did not always agree. DeepFilterNet3 gave the largest SNR improvement and usually the highest STOI, but it also caused more substitutions and insertions. This supports the general concern in [8], [9], although the experimental settings differ. Plantinga et al. improved ASR by training the enhancement loss with acoustic-model information [8]; the present pipeline has no such coupling. Braun and Gamper found that extra pretrained loss networks did not automatically improve ASR over a strong spectral loss [9]. The present results show a similar suppression-distortion trade-off for fixed Whisper base.en, but they do not establish the same mechanism or a result that applies to all ASR systems.

The alpha=1.5 change did not improve the independent test result. It gave the lowest WER on the five development utterances and increased output SNR. However, its test WER was worse than the original alpha=1.0 setting. The small development set may have contributed to an unstable parameter choice. The result also shows that stronger noise removal does not always keep the features used by Whisper.

Several limitations restrict generalisation. The 50 utterances came from a project-specific LibriSpeech development partition and represented read English only. Only one channel per DEMAND recording and one Whisper size were tested. STOI is an intelligibility predictor rather than a listening test, and PESQ/MOS were not measured. The DeepFilterNet binary was treated as a fixed pretrained system, so architecture-specific causes of the ASR mismatch were not isolated. Finally, repeated conditions from each utterance are correlated; clustered intervals address this partly, but a larger held-out corpus would be preferable.

6. Conclusion

For the fixed Whisper base.en recogniser, enhancement helped in some conditions but not consistently. Original spectral subtraction gave the best overall WER, 10.25% compared with 10.53% without enhancement, but the 0.27-point reduction was uncertain. Wiener filtering and the stronger spectral-subtraction setting did not improve overall WER. DeepFilterNet3 increased output SNR and improved STOI in most conditions, but it increased WER to 14.30%. This directly

answers the project question: better signal scores can occur together with worse recognition.

My work was to extend the original MATLAB experiment into a complete MATLAB-Python study. I prepared common noisy mixtures, implemented the STFT methods, connected DeepFilterNet3 and Whisper, and calculated WER, STOI, SNR, log-Mel distance, and runtime. I also tested one change to spectral subtraction and checked all 2,300 recognition inputs. The external resources were LibriSpeech, DEMAND, OpenAI Whisper, DeepFilterNet3, jiwer, and MATLAB's STOI function. No neural model was trained. Future work could use more speakers, more Whisper sizes, more changing noise types, and a larger development set.

7. References

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